Example: Determine

$$\int \frac{1}{x^2 - a^2}\,dx$$

for $a \neq 0$.

How do we determine what A and B are? cross multiply

$\frac{1}{x^2-a^2} = \frac{1}{(x-a)(x+a)} = \frac{A}{x-a} + \frac{B}{x+a}$

completing the square

Splitting a product into a sum. But to add fractions they need to have a common denominator

these are fancy 1's

$\frac{A}{x-a}\frac{(x+a)}{(x+a)} + \frac{B}{(x+a)}\frac{(x-a)}{(x-a)}$

$1 = A(x+a) + B(x-a)$

If $x = a$: $1 = 2Aa$, $A = \frac{1}{2a}$

If $x = -a$: $1 = B(-2a)$, $B = \frac{-1}{2a}$

$\int \frac{1}{x^2-a^2}\,dx = \frac{1}{2a}\int \frac{1}{x-a}\,dx - \frac{1}{2a}\int \frac{1}{x+a}\,dx$

$u = x-a$, $du = dx$; $u_2 = x+a$, $du_2 = dx$

$= \frac{1}{2a}\int \frac{1}{u}\,du - \frac{1}{2a}\int \frac{1}{u_2}\,du_2$

$= \frac{1}{2a}\ln|u| - \frac{1}{2a}\ln|u_2| + C$

$= \frac{1}{2a}\ln|x-a| - \frac{1}{2a}\ln|x+a| + C$

Partial Fraction Decomposition: A rational function $f(x)$ takes the form

$$f(x) = \frac{p(x)}{q(x)},$$

where $\text{degree}(p(x)) < \text{degree}(q(x))$. To integrate, it is often easier to express a rational function as a sum of simpler rational functions:

$$f(x) = f_1(x) + f_2(x) + \cdots + f_k(x),$$

where each $f_i(x)$ takes the form

$$\text{Linear}\ \frac{A}{(ax+b)^n} \quad \text{or} \quad \text{quadratic}\ \frac{Ax+B}{(ax^2+bx+c)^n}.$$

Given a rational function $f(x)$ of form $f(x) = \frac{p(x)}{q(x)}$ with polynomials $p(x)$ and $q(x)$ — degree of $q(x)$ has to be bigger than degree of $p(x)$. Decompose $f(x) = f_1(x) + f_2(x) + \dots f_k(x)$ where $f_i(x) = \frac{A}{(ax+b)^n}$ (Linear) or $f_i(x) = \frac{Ax+B}{(ax^2+bx+c)^n}$ (quadratic)

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Example: Find

$$\int \frac{x+2}{(x^2+2x+1)(x-1)}\,dx.$$

$\frac{x+2}{(x^2+2x+1)(x-1)} = \frac{x+2}{(x+1)^2(x-1)}$

$\frac{x+2}{(x+1)^2(x-1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-1}$

Every iteration — if it was $(x+1)^3$ you'd have $x+1$, $(x+1)^2$, and $(x+1)^3$. You want every power up to the one represented

why this the cross mult?

To get common denominator: $x+2 = A(x+1)(x-1) + B(x-1) + C(x+1)^2$

Now solve for A, B and C. Choices need to hold for any val of x

If $x = 1$: $3 = C4$, $C = \frac{3}{4}$

If $x = -1$: $1 = -2B$, $B = -\frac{1}{2}$

If $x = 0$: $2 = -A - B + C$, $2 = -A + \frac{1}{2} + \frac{3}{4}$, $A = -\frac{3}{4}$

once you solve for these they are a function of x but these are constants that will hold for any value x

$-\frac{3}{4}\int \frac{1}{x+1}\,dx - \frac{1}{2}\int \frac{1}{(x+1)^2}\,dx + \frac{3}{4}\int \frac{1}{x-1}\,dx$

$u = x+1$, $du = dx$; $u_2 = x-1$, $du_2 = dx$

$-\frac{3}{4}\int \frac{1}{u}\,du - \frac{1}{2}\int \frac{1}{u^2}\,du + \frac{3}{4}\int \frac{1}{u_2}\,du_2 = -\frac{3}{4}\ln|x+1| + \frac{1}{2}\frac{1}{x+1} + \frac{3}{4}\ln|x-1| + C$

What happens when degree$(p(x)) \geq$ degree$(q(x))$?

Example: Find

$$\int \frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1}\,dx.$$

Degree of top is 3 and bottom is 2 so this breaks the rule! So we need to get this in a workable form first

Polynomial long division

$$\begin{array}{r l} & x - 6 \\ x^2 + 0x - 1 \,\big) & x^3 - 6x^2 + 5x - 3 \\ & -(x^3 + 0x^2 - x) \\ \hline & -6x^2 + 6x - 3 \\ & -(-6x^2 + 0x + 6) \\ \hline & 6x - 9 \end{array}$$

Multiplying this + adding remainder (divided by denom) gives a function equivalent to OG function

$6x-9 \leftarrow$ " remainder "

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$$= \overbrace{\int (x-6) + \frac{6x-9}{x^2-1}\,dx}^{\text{= via PLD}} = \underbrace{\int x+6\,dx}_{\text{This can easily be integrated}} + \int \frac{6x-9}{x^2-1}\,dx$$

$$\int \frac{6x-9}{(x+1)(x-1)}\,dx = \int \frac{A}{x+1}\,dx + \int \frac{B}{x-1}\,dx = A\ln|x+1| + B\ln|x-1| + C$$